Information System of Smart Hospital

Computer and Communications Center (IT Department),
Taichung Veterans General Hospital (TCVGH), Taichung, Taiwan

Present by: Acting Director Huang, Sheng-Yang
黃勝揚

Date: June 5th, 2026

Location: Taichung, Taiwan

Taichung Veterans General Hospital

- **Established in 1982**
- **The only Government funded National Medical Center in Central Taiwan Area**

臺中榮民總醫院
Taichung Veterans General Hospital

Taichung Veterans General Hospital

Tertiary-Level Medical System in Central Taiwan Area

1 **Medical Center**

3 **Regional Hospitals**

4 **Veteran Home Clinics**

Total **8** **Hospitals and Clinics**

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Taichung Veterans General Hospital

Taichung Veterans General Hospital (2024)

Total: >1600 beds

>3.0 M Registered Patients from Taiwan and other countries

Veterans: <10%, public: >90%

10,472
Outpatient Visits / Day
(2.78 M+ /year)

210
Emergency Patients / Day
(76K+ / year)

6,813
Inpatients / Month
(81K+ / year)

5,054
Operations / Month
(60K+ / year)

Average Length of Stay: 5.11 Days
Shortest LOS in Taiwan Medical Centers

Agenda

- Important Awards and Achievements
- Hospital Information System
- Smart Hospital
- Future Plan

Important Awards and Achievements

臺中榮民總醫院
Taichung Veterans General Hospital

International Recognition

Newsweek

The only hospital in Taiwan listed in
Top 100 World's Best Smart Hospitals in 2025 and 2026.

Top 1 Hospital in Taiwan in 2023-2026.

Newsweek Nation World Lifestyle Opinion

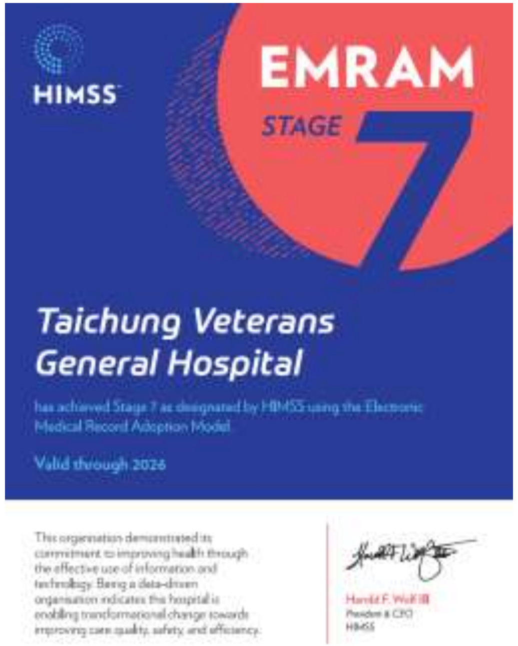
Rank	Hospital	City	Country
83	Rigshospitalet - Copenhagen University Hospital	Copenhagen	Denmark
84	AdventHealth Orlando	Orlando	United States
85	Taichung Veterans General Hospital	Taichung	Taiwan
86	Policlinico Universitario A. Gemelli	Rome	Italy

臺中榮民總醫院
Taichung Veterans General Hospital

International Recognition

HIMSS EMRAM Stage 6 and 7 Certification (2023)
The Fastest Record in Taiwan Hospital's History (within 8 months)

The Second Highest Global Ranking of DHI (Digital Healthcare Indicator) Score (2024)



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Taichung Veterans General Hospital

International Recognition

HIMSS

HIMSS Davies Award (2025)
1 of 4 hospitals worldwide


DIGITAL HEALTH

Taichung Veterans General Hospital Earns 2024 Davies Award for Using AI to Improve Care, Safety and Reduce Pain

Published on October 1, 2024

Hospital Information System

臺中榮民總醫院
Taic!

Hospital Information System Overview

-Patient Web Portal
-Patient APP
-Remote Healthcare
. . . . .

Patient Engagement

Enhance Patient Engagement & Satisfaction

Healthcare Team Collaboration

Improve Health Team Performance

-Outpatient / Inpatient / Emergency System
-Traditional Chinese System
-CPOE System
-ENote System
-Nursing System and APP
-PACS / RIS System
-LIS System
-Pathology System
-Pharmacy System
-OR System
-ANES System
-PCU System
-Dietary System

Data Integration & Analysis

Data-Driven Decision

-HR System
-Accounting System
-Purchasing System
-Stock Management System
-Payment System
-Medical Device Management System

Hospital Administration

Enhance Support Team Performance

-Data Warehouse
-Dashboard System
-OMOP
-TriNETX
-KM System

Teaching And Researching

Enhance Teaching and Research Quality

-E-Learning System
-E-Portfolio System
-Professional Certificate Management System
-Research Management System
-Clinical Trial System

First Innovations of Information System in Taiwan

- (2005) First hospital – Mobile Nursing Carts with Web-based Nursing Information System fully implementation
- (2009) First hospital – legally implemented EMR
- (2014) First hospital – integrate with National Health Insurance (NHI) Medi-Cloud system
- (2018) World's first hospital - E-Paper with UniDose Cartridge
  (2019) World's first hospital - E-Paper with Drip Infusion Label
- (2018) First hospital - Smart Operation Room Management System
- (2022) Largest Telehealth Care Center
- (2025) First hospital exchange FHIR with NHIA

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Taichung Veterans General Hospital

Digital Hospital Journey - Paperless

First hospital legally implemented EMR (2009) → **Fully paperless EMR (2016)** → **Scan historical paper medical records to E-files (2016~)** → **Patient inform /consent forms (2021)**

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Taichung Veterans General Hospital

Digital Hospital Journey - Filmless

2000
First fully filmless hospital in Taiwan

2018
Pathology slide digital image system

2019
Video and VNA image system

2020
Patient provide non-TCVGH images system

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Taichung Veterans General Hospital

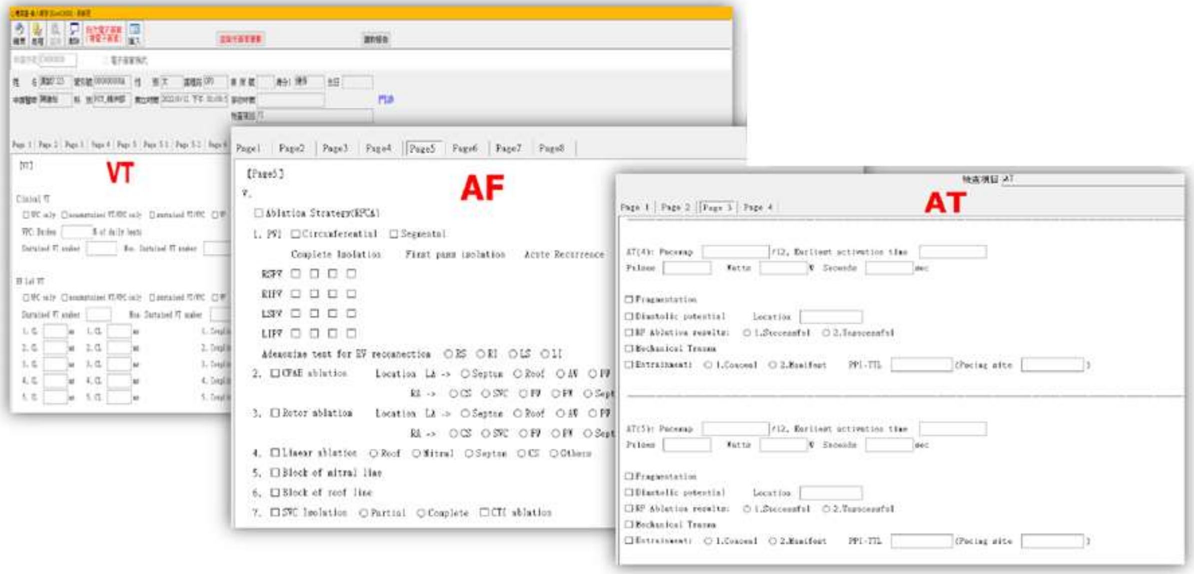
Smart Hospital - Structured Medical Record

Structured ADM/ DIS/ Progress/ OR/ Nursing Records (2004)

Structured Examination Reports (2022)

Structured Outpatient SOAP (2023)

Structured Outpatient PROM (2023)



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Taichung Veterans General Hospital

Smart Hospital - Standardization of Medical Record

FHIR Connection (2020~)

First hospital proposed FHIR format (2022)

First hospital exchange FHIR with NHIA (2025)

醫師科	臨床科別	FHIR送件數量	XML送件數量	合計	FHIR上傳率
GS	一般外科	15	0	15	100.0%
DCM	胸腔部	145	54	199	72.9%
HEMA	血液腫瘤科	32	0	32	100.0%
GI	胃腸肝膽科	24	0	24	100.0%
BS	乳房腫瘤外科	76	1	77	98.7%
NS	神經外科	23	9	32	71.9%
ONCO	腫瘤醫學中心	105	35	140	75.0%
META	內分泌新陳代謝科	5	3	8	62.5%
GU	泌尿醫學部	42	52	94	44.7%
CRS	大腸直腸外科	13	18	31	41.9%
GO	婦癌科	4	0	4	100.0%
CV	心臟內科	0	2	2	0.0%
GYN	婦科	12	1	13	92.3%
ROD	放射腫瘤科	1	0	1	100.0%
PED	兒童醫學部	2	11	13	15.4%

Among the highest percentage of FHIR cancer therapy pre-approval review in Taiwan

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Taichung Veterans General Hospital

Smart Pharmacy Management System

No.1 Hospital in Taiwan Integration with NHIA Medi-Cloud
Patent donated to NHIA, every hospital in Taiwan got this benefit

Green Book
Indicates
Medication from HIE (Other Hospital / Clinic)
And Link to TFDA Web Page

Red Book
Indicates
Medication from Taichung Veterans General Hospital

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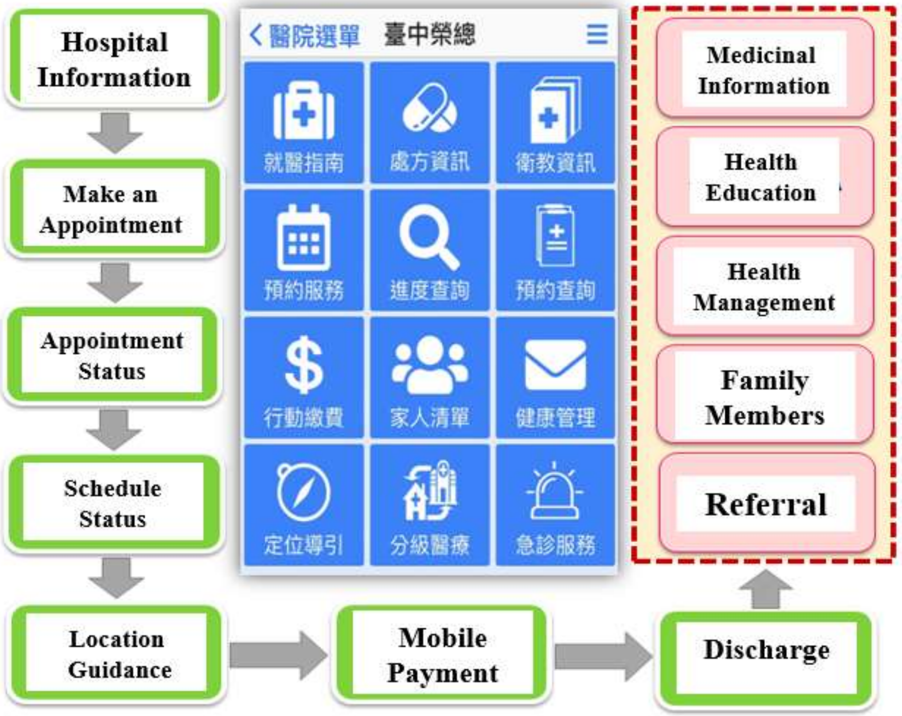
Taichung Veterans General Hospital

Smart Patient Engagement Systems



Patient APP

Personal Health Data and Trend Analysis (also include family)

Patient Self Report Outcome

- Provide Full Function for Inpatient / Emergency / Outpatient Patients
- Provide for General Practitioner Clinics

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Taichung Veterans General Hospital

Deploy Information System to 12 Veteran Hospitals around Taiwan

Fastest record in Taiwan Hospital's History
One of Largest Hospital Information Systems in Taiwan

Impossible Missions
- During COVID Period (2019-2020)
- 8 Hospitals in 8 months

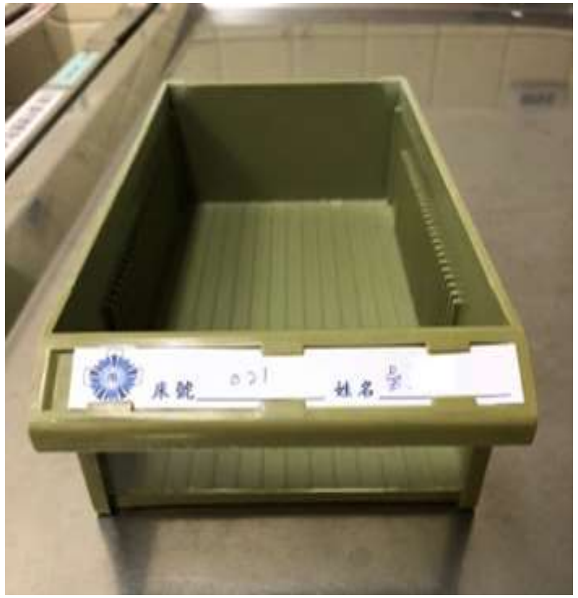
Smart Hospital

臺中榮民總醫院
Taichung Veterans General Hospital

E-Paper Systems

- World's 1st and Largest Successful Story (since 2018)
- Combines E-Paper with UniDose Cartridge (integrate with PIS)

Outcome: save up to 60 minutes each day (each pharmacist)



Before

After

臺中榮民總醫院
Taichung Veterans General Hospital

E-Paper Systems

- World's 1st and Largest Successful Story (since 2019)
- Combines E-Paper with Drip Infusion Label (integrate with NIS)

Outcome: save up to 60 minutes each day (each nurse)

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Taichung Veterans General Hospital

Smart Operation Room Management System

Scheduling → Operation Room → Dashboard

Outcomes

- Increase >3800 operation hours each year (equal to 1 new operation room)
- Increase Patient Safety

In case of 150% over expected surgical duration, OR will initiate a warning process to ensure patient safety.

Tele Healthcare Center

- **Taiwan's largest Tele Healthcare Center** (since 2022)
- **IoT-5G technology connect 25+ locations**

  18 partner hospitals
  4 Veterans Homes
  3 long-term care institutions

- **5G + X (all are 1st in Taiwan)**

5G + AR Glass

5G + Cardiac Catheterization

5G + Da Vinci

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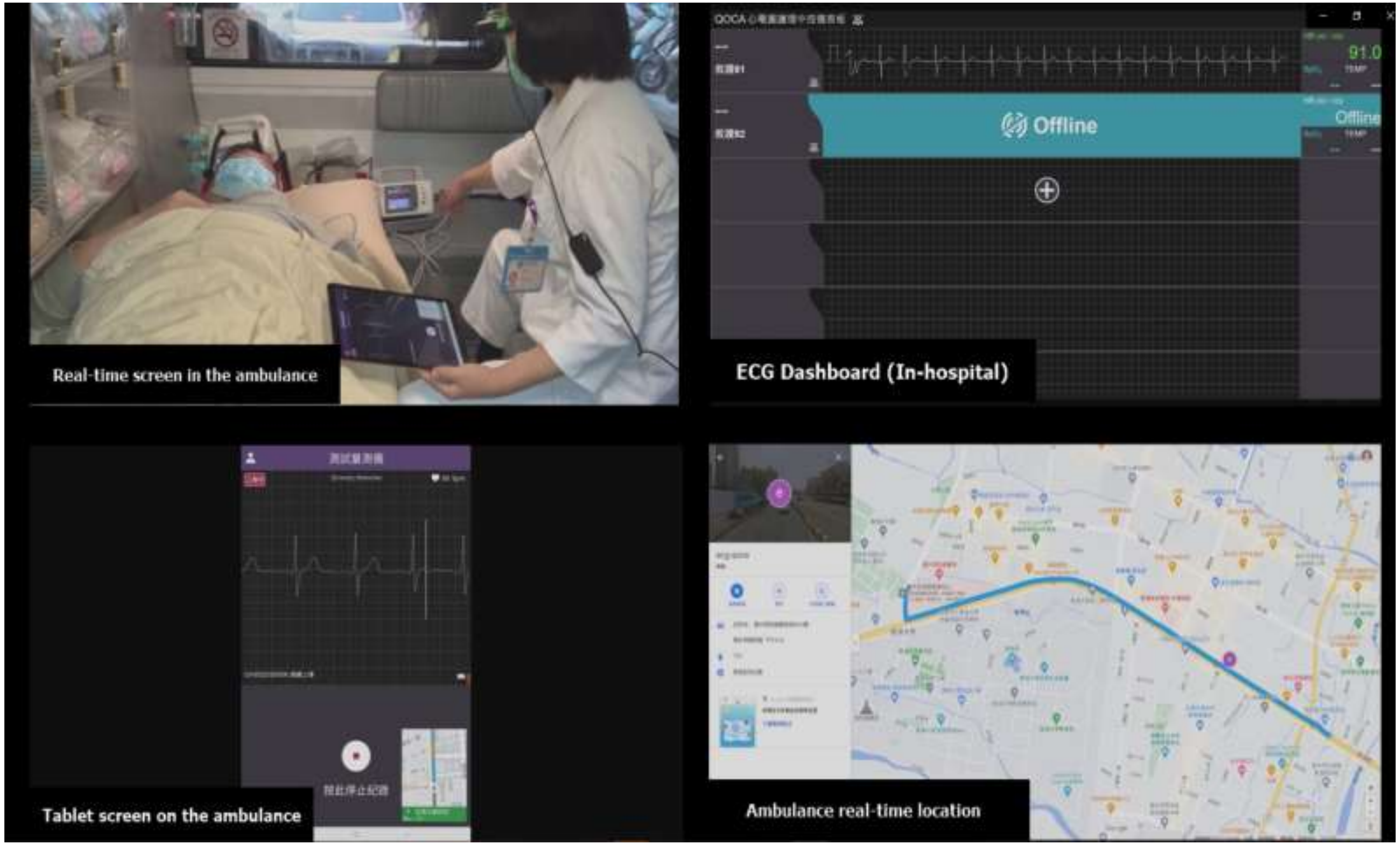
Taichung Veterans General Hospital

Remote Collaboration among Medical Professionals

In Chiayi Branch Hospital

Remote Consultation



Real-time vital sign monitoring in Ambulance car

https://www.youtube.com/watch?v=cX1qFdCydeQ

Taiwan's Largest Fully Automatic Laboratory System (since 2023)

- Emergency Specimen LAB Results

  < 60 mins

- Laboratory Test Professionals

  25% reduced

AI and LLM

SaMD and LLM
all running on local servers

TFDA Approved SaMD -
AI for Early Detection of Osteoporosis (VeriOsteo) (2023.12)

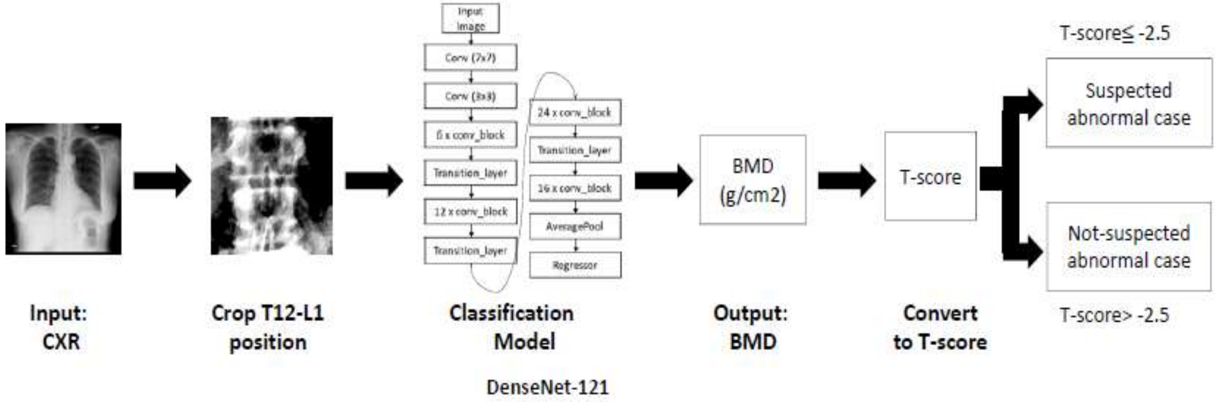
VeriOsteo® OP: Taiwan's First Smart Medical Device

Using Deep Learning Technology for Early Prevention of Osteoporosis

Product Introduction

VeriOsteo® OP is an innovative medical product that utilizes deep learning technology to predict bone density and the risk of osteoporosis through standard chest X-rays. It offers a novel approach to provide early warnings in the initial stages of osteoporosis before the necessity of traditional DXA (Dual-Energy X-ray Absorptiometry) scans.

This breakthrough tool is set to revolutionize the way we approach the prevention and treatment of osteoporosis.




衛生福利部醫療器材許可證

衛部醫器製字第 007973 號

中文品名："宏碁智醫" 智骨篩-骨質密度異常篩檢軟體

英文品名："Acer Medical" VeriOsteo OP

類　　別：第P類：放射學科學

醫療器材商名稱：宏碁智醫股份有限公司

規格或型號：VeriOsteo OP-01

以下空白

製造業者名稱：宏碁智醫股份有限公司

製造業者地址：新北市汐止區新台五路1段110號24樓

效能、用途或適應症：詳如核定之中文說明書

前項醫療器材經本部審核與醫療器材管理法之規定

衛生福利部部長

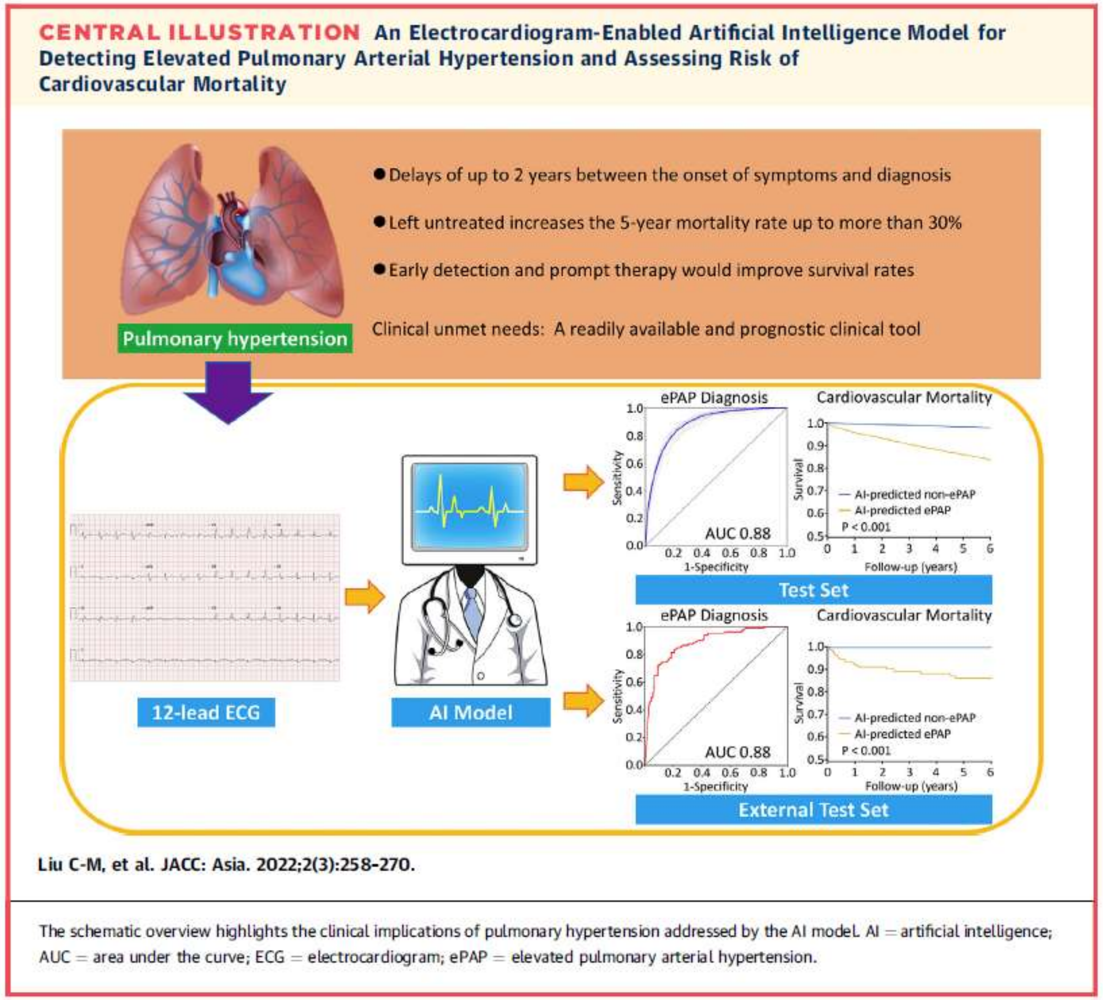
薛瑞元

MF 017437

TFDA Approved SaMD -
AI for Acute Kidney Injury Prediction (2023.12)

Acura AKI人工智慧急性腎損傷預測輔助軟體

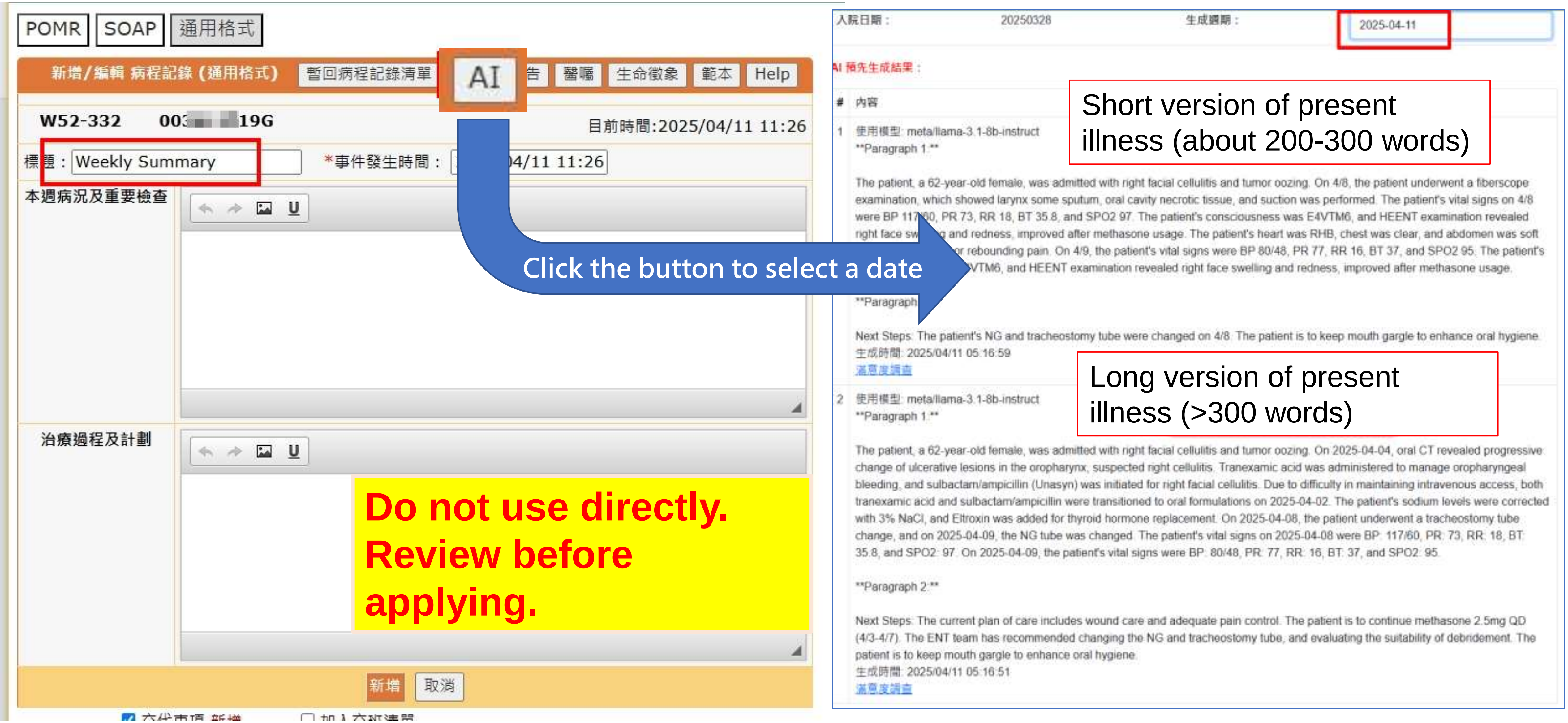
TFDA Approved SaMD - AI Prediction of Mortality With Pulmonary Hypertension (based on ECG) (2025.3)



The schematic overview highlights the clinical implications of pulmonary hypertension addressed by the AI model. AI = artificial intelligence; AUC = area under the curve; ECG = electrocardiogram; ePAP = elevated pulmonary arterial hypertension.


衛生福利部醫療器材許可證
衛部醫器製字第008156號
製造許可編號：QMS1894
中文品名："台灣醫學影像" 肺高壓偵測系統
英文品名：TAIMedImg PHDS
醫療器材商名稱：台灣醫學影像股份有限公司
規格或型號：詳如核定之中文說明書
製造業者名稱：台灣醫學影像股份有限公司
製造業者地址：臺北市大同區承德路1段17號13樓
效能、用途或適應症：詳如核定之中文說明書
衛生福利部部長
邱泰源
MF 019001

LLM - Generation of Weekly Summary / Admission Summary / Discharge Summary / Others …



Future Plan

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Taichung Veterans General Hospital

Future Plan

Develop Next-Generation Hospital Information System

Adopt smart technologies in hospital
AI, Big Data, IOT, Tele healthcare, Image, NLP, LLM, Robotics, Digital Twin, …

Network Security & Privacy Protection

Cooperate with cloud provider

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Taichung Veterans General Hospital

Conclusion

- High Table Support and Multiple Discipline Cooperation are Keys to Success
- People and Process First, Technology last
- Good data makes good results
- Cooperation with Hospitals, Universities and Vendors

Thank You

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